BG-NASEMU-2026-11-13 — Research report (rev 2, primary-source pass)

Retrieval date: 2026-11-13. Rev 2 replaces the shallow rev 1. Method: GitHub REST API (`gh api`, authoritative for licence SPDX / pushed_at / latest release), primary README fetches (vdsm/virtual-dsm, dockur/windows, dockur/macos, drakkan/sftpgo, AuxXxilium/arc, samba-container, Garage S3-compat page), three extended web searches, local host probes, and a read of the Containers branch `fix/containers-g1-g2-g11` (clone `/home/milosvasic/.cache/claude-tmp/cg.XKZDrl/c`, read-only). No image pulled, no VM booted, no credentials touched. "Last push" = repo `pushed_at` (activity signal, not a release). UNCONFIRMED = not verified from a primary source.

0. Host facts (measured 2026-11-13)

- `id`: uid 1000, NOT in group `kvm`; `getfacl /dev/kvm` shows `user:milosvasic:rw-`; `[ -r -w /dev/kvm ]` => OK. **KVM is usable by this user via ACL** (the Containers `hostbudget.KVMUsable` doc confirms the pitfall: stat is not enough, open O_RDWR). Note: ACL grants can disappear on reboot/udev events — re-probe each run.
- 16 CPUs, 30 GiB RAM (about 16 GiB available, 3 GiB free + 13 cache), `/` 1.8T 97% used, 57 GiB free (also /home); extra disks: `sda1` 7.3T (mountpoint not shown/mounted? UNCONFIRMED — candidate for VM disks, needs operator decision), `nvme0n1` 931G (partitions unmounted per lsblk output, UNCONFIRMED purpose; do not touch). Load avg ~18-23 (other stacks: many `helixagent-*`/`helixcode-*` containers plus catqa stacks running).
- Podman 5.7.0 rootless, netavark; `pasta` present; `ip_unprivileged_port_start=1024` (rootless cannot bind 445/21/22/2049/111 without remap or sysctl change). Tools present: smbclient, showmount, curl, sftp. Local images: `cat-testservers:1/2/3`, nginx:alpine, debian slim.
- **Key finding: Catalogizer already ships a lite tier.** `docs/qa/test-servers/up.sh` (+ Containerfile, smb.conf, ftps_server.py, verify.sh, ports.env) runs rootless containers `cattsrv-smb` (smbd), `-ftp`, `-ftps` (python), `-sftp`, `-webdav`, `-http`, `-nfs` using **`rclone serve <proto>`** over a seeded corpus (`/home/milosvasic/Testing-servers/corpus`), with `--network host`, LAN-IP auto-detect, self-signed TLS, credentials from a 0600 env file. Gaps vs. the mandate: lives in the app repo (not reusable Containers code), `--network host` (port clash risk), no golden-manifest evidence JSON, no S3, no VM tier, no hostbudget admission. The new work therefore GENERALISES this into Containers rather than starting from zero.

1. Containers submodule facts (branch fix/containers-g1-g2-g11, NOT yet upstream)

Commits on the branch: `a8e8c48` G1 typed run options, `cd782c0` G2 hostbudget + admission before emulator boot, `4db2931` G11 reachable-endpoint resolver with in-target probes.

- `pkg/hostbudget` EXISTS only on this branch (~660 lines): closed verdicts `fits | does_not_fit | undetermined`; `Need{MemBytes,CPUs,Pids,NeedsKVM}`; `Policy` (defaults: 60% mem ceiling, reserve max(4 GiB,15%), load/CPU 0.75, mem PSI some>10 / full>2, cpu PSI some>50); `Probe` injectable seams (ProcRoot, KVMPath, PidsLimit); `Admit(need,policy,probe) Evidence`; `Evidence.WriteJSON(dir,label)`; `KVMUsable(path)`. Docs say PROVISIONAL: thresholds not calibrated on a long real run. **Gap: no disk-free axis** (needed: host `/` is 97%).
- `pkg/runtime` (branch): typed `RunOptions` `WithRunMemory/MemorySwap/CPUs/PidsLimit/Device/Userns("keep-id")/Publish(hostIP:hostPort:ctr/proto)/...` with digest and label options and refusal errors (`ErrRunForbiddenOption` for host-escalating forms, `ErrRunUnsupportedOption`, `ErrRunInvalidOption`). Runtimes: docker, podman, crio, nerdctl, lxd, kubernetes. Host-escalating options (privileged, host net) are refused by design — which conflicts with the current `--network host` test-servers; a nasemu design must use `Publish` or a pasta/macvlan path.

- `pkg/emulator`: `Containerized` (Boot/WaitForBoot/Teardown, Android oriented), `buildContainerRunArgs[WithLimits]`, `defaultEmulatorLimits`, `kvmAvailable`. `pkg/vm`: `qemu.go`, `network.go`, `matrix.go`, `teardown.go`, `macos/` (`RunOptions`), `ios/`. `pkg/health`: http/tcp/grpc/custom checkers. `pkg/network`: ephemeral/overlay. These are reusable for the VM tier; a NAS profile layer would sit above runtime+health+hostbudget.

2. Candidate catalogue (primary-source verified unless marked)

Item	URL	Licence (primary)	Maintenance (GitHub API 2026-11-13)	Resources	Rootless podman	Network	Legal / pinning
Virtual DSM	https://github.com/vdsm/virtual-dsm ; image `vdsm/virtual-dsm`	MIT (repo)	latest release v7.67 published 2026-11-12; pushed 2026-11-12; not archived; ~4k stars	readme: >=1 GB RAM, >=16 GB disk, defaults 2 CPU/2 GB (`CPU_CORES`,`RAM_SIZE`); Docker/Podman on Linux with KVM; not Docker Desktop	Podman listed as supported; **no rootless guidance in readme (UNCONFIRMED whether tun/`NET_ADMIN`/vhost-net work rootless)**	bridge/NAT default (shares host IP), macvlan (own IP, host can't reach it without 2nd macvlan), DHCP mode needs `/dev/vhost-net` + device_cgroup_rules	FAQ: code is open-source, no Synology material shipped; installing requires accepting Synology EULA which **does not permit non-Synology hardware; "only run on an official Synology NAS"**. DSM `.pat` auto-downloaded (default DSM 7.0.1) or supplied via `URL` / bind `/boot.pat` => supply a pinned local `.pat` for reproducibility; its sha256 can be recorded. Pin container by digest.
XPEnology: Arc loader (AuxXxilium/arc), RR (RROrg/rr), redpill-lkm	https://github.com/AuxXxilium/arc ; https://github.com/RROrg/rr ; https://github.com/RedPill-TTG/redpill-lkm	arc: GPL-3.0 (API) but README says "commercial use is not permitted and strictly forbidden"; RR GPL-3.0; redpill-lkm GPL-3.0	arc 4.2.3 published 2026-11-01; RR pushed 2026-09-18; redpill-lkm last push 2022-10-22 (stale, not archived)	x86-64 machine/VM you own, 4 GB loader disk; Docker alternative runs Arc in QEMU	UNCONFIRMED	VM-level	README: DSM owned by Synology; educational use. Bootloader hack that boots DSM off non-Synology hardware: same EULA problem as vdsm plus circumvention connotation. **Recommend exclude**; vdsm is the cleaner path (same EULA issue, no loader hack).
dockur/windows	https://github.com/dockur/windows	MIT	v6.06 published 2026-10-05; pushed 2026-11-12; not archived; 54k stars	>=2 GB RAM, >=32 GB disk, KVM; default 2 CPU/4 GB; compose uses `/dev/kvm`; readme suggests `privileged: true` only to debug	Podman supported; no rootless guidance in readme (UNCONFIRMED)	NAT default; macvlan; DHCP; ports 8006 (web viewer), 3389 TCP/UDP	Project ships no Windows; user must hold valid licence; keys in code are Microsoft generic trial keys.
dockur/macos	https://github.com/dockur/macos	MIT	v3.12 published 2026-08-21; pushed 2026-11-12; not archived	AVX2 CPU, >=4 GB RAM, >=32 GB disk, default 64 GB disk, 2 CPU/4 GB; AMD: <=8 GB during install	UNCONFIRMED	ports 8006, 5900 (VNC)	Readme: "Only run this container on Apple hardware, any other use is not permitted by their EULA." Highest legal risk. (Upstream OSX-KVM kholia GPL-3.0 push 2025-11-11; sickcodes/Docker-OSX GPL-3.0 push 2026-01-26; not required.)

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Samba (samba-in-kubernetes)	https://github.com/samba-in-kubernetes/samba-container ; images `quay.io/samba.org/samba-server`	GPL-3.0	v0.9 published 2026-06-18; pushed 2026-11-11; ~98 stars (small project, Samba-team backed)	small	**Documented**: example maps host 10445->ctr 445 when rootless; file-server image needs no special privilege; AD-DC image needs `--privileged` (irrelevant)	publish only	Config via `SAMBACC_CONFIG` JSON; default user `sambauser` / password `samba` (change it; never use default on LAN). Existing Catalogizer approach (smbd in alpine image with smb.conf) also works rootless. Pin by digest.
ServerContainers/samba	https://github.com/ServerContainers/samba	licence not reported by API	pushed 2026-07-11, no releases	small	UNCONFIRMED	—	Alternative; lower priority. dperson/samba is stale (Docker Hub ~5 y).
rclone serve (ftp, sftp, webdav, http, s3, nfs, dlna)	https://github.com/rclone/rclone	MIT	v1.76.0 2026-11-06; pushed 2026-11-12	tiny, single Go binary	fine (high ports)	publish	**Already used by Catalogizer test-servers incl. `serve nfs`.** Fidelity caveat: rclone is a faithful-enough protocol faker, not a real vsftpd/OpenSSH/Apache; real-server quirks are not exercised.
SFTPGo	https://github.com/drakkan/sftpgo	AGPL-3.0 with additional terms in NOTICE; WebAdmin/WebClient theme proprietary (primary readme)	v2.7.6 published 2026-09-19 (rev 1 said v2.7.5, superseded); pushed 2026-11-09	small	fine	SFTP, FTP/S, WebDAV, HTTP; S3 is only a storage backend, not a server protocol	Community vs Enterprise split; Enterprise is faster-moving. OK as unmodified test infra; record in dependency register.
vsftpd	upstream is scarif.net/security.appspot (not GitHub; GitHub path 404)	GPLv2 (general knowledge; UNCONFIRMED here)	UNCONFIRMED	tiny	needs port remap; passive range	—	distro package; fidelity candidate for "real FTP server"
ProFTPD	https://github.com/proftpd/proftpd	GPL-2.0	v1.3.9e 2026-10-26; pushed 2026-11-12	small	ok with remap	—	real FTP server alternative
Garage (S3)	https://github.com/deuxfleurs-org/garage ; docs https://garagehq.deuxfleurs.fr/documentation/reference-manual/s3-compatibility/	AGPL-3.0	no GitHub Releases; tags v2.5.0/v2.4.1/v2.4.0; pushed 2026-11-07 (git.deuxfleurs.fr itself blocked by bot-protection for fetch)	small	fine	S3 3900	S3 compat page: SigV4 yes, SigV2 no, ListObjectsV2 yes, all 7 multipart ops yes, **object versioning NOT supported** (GetBucketVersioning stub). Operator memory notes Garage already chosen for S3.
SeaweedFS	https://github.com/seaweedfs/seaweedfs	Apache-2.0	4.53 published 2026-11-08; pushed 2026-11-12; 36k stars	moderate (master+volume+filer+s3 or `weed server -s3`)	fine	S3, WebDAV, FUSE, NFS-ish filer	Permissive alternative to AGPL S3; versioning support UNCONFIRMED.

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MinIO	https://github.com/minio/minio	AGPL-3.0	**archived**, last push 2026-04-24, last release RELEASE.2025-10-15 (no community binaries/images since); community fork pgsty/minio AGPL-3.0 with release 2026-11-12	—	—	—	Exclude (upstream dead). pgsty fork UNCONFIRMED quality/ trust.
Ceph RGW	https://github.com/ceph/ceph	mixed (API NOASSERTION)	active	very heavy	impractical	—	Exclude (overkill for the host).
NFS-Ganesha	https://github.com/nfs-ganesha/nfs-ganesha	LGPL (API NOASSERTION; UNCONFIRMED)	V15.11 2026-11-07; active	small	**Support-list thread (2022): VFS FSAL under rootless Podman fails with `vfs_open_by_handle` EPERM; one participant said rootless VFS is "pretty much not possible"; no confirmed working setup.** Treat as NOT feasible rootless; needs CAP_DAC_READ_SEARCH-class capabilities.	2049 + rpcbind/ mountd	Do not plan on it.
unfs3	https://github.com/unfs3/unfs3	NOASSERTION (BSD-like, UNCONFIRMED)	v0.11.0 2025-03-28; push 2025-07-03	tiny	userspace NFSv3; rootless plausible, UNCONFIRMED	2049/mountd	Backup NFS candidate; NFSv3 only.
`rclone serve nfs`	(rclone)	MIT	see rclone	tiny	works rootless (already in Catalogizer test servers)	custom port + `mount -o port=,mountport=`	Best-evidenced NFS path for lite tier (NFSv3). Kernel nfsd needs privileged — excluded.
dufs	https://github.com/sigoden/dufs	Apache-2.0	v0.47.0 2026-10-17; push 2026-11-10	tiny	fine	HTTP + WebDAV	WebDAV alternative.
hacdias/webdav	https://github.com/hacdias/webdav	MIT	v5.17.3 2026-10-30; push 2026-11-11	tiny	fine	WebDAV	Alternative. Apache `mod_dav` / nginx-dav: well-known, not individually verified (UNCONFIRMED).
OpenMediaVault	https://github.com/openmediavault/openmediavault	GPL-3 (API NOASSERTION)	pushed 2026-11-09	full Debian NAS OS	VM-only, no container	—	Heavy; optional VM-tier alternative, not needed.
TrueNAS (truenas/ middleware)	https://github.com/truenas/middleware	LGPL-3.0	pushed 2026-11-12; no GitHub Releases	full OS, ZFS	VM-only	—	Optional VM-tier alternative; not needed.
CasaOS	https://github.com/IceWhaleTech/CasaOS	Apache-2.0	v0.4.15 2024-12-19 (stale releases)	home-server UI	n/a	—	Not a NAS protocol emulator; exclude. Unraid: proprietary OS, not researched (UNCONFIRMED); exclude.

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Synology Web API docs	https://help.synology.com/developer-guide/integrate_dsm/web_authentication.html ; File Station API Guide PDF `cndl.synology.cn/download/Document/Software/DeveloperGuide/Package/FileStation/All/enu/Synology_File_Station_API_Guide.pdf`	proprietary docs	—	—	—	—	URLs located; PDFs not opened this pass (UNCONFIRMED parameter details). Login via `SYNO.API.Auth`; 2FA error 406 reported by community only.
Open-source Synology API mocks	none found	—	—	—	—	—	Search found only clients (py-synologydsm-api, kwent/syno, ProtoThis/python-synology) and an MCP server whose tests use a "fake DSM over real HTTP" (joaomelorodrigues/synology-mcp, UNCONFIRMED coverage). **No standalone reusable DSM mock exists** => a FileStation/Auth shim would be our own work; real fidelity only from vdsm.

3. Conclusions

1. Lite tier: reuse the proven Catalogizer approach (rclone serve ftp/sftp/webdav/http/nfs + real smbd) and generalise it into Containers; add Garage (or SeaweedFS, Apache-2.0) for S3; add a "real server" fidelity option (ProFTPD/vsftpd) later. Kernel NFS and NFS-Ganesha are not rootless-feasible; `rclone serve nfs` is the evidenced path.
2. VM tier: vdsm (MIT, very active, v7.67 yesterday) is the only sensible DSM route; its EULA problem (Synology hardware only) is explicit in the project's own FAQ and cannot be mitigated technically — operator attestation is the only gate. XPEnology loaders add circumvention connotation: exclude. macOS: project itself restricts to Apple hardware: opt-in only. Windows: operator-supplied licence.
3. Host: KVM works for this user via ACL, but RAM headroom (~16 GiB), 57 GiB free disk, and load ~20 make the VM tier admit-or-refuse material; hostbudget lacks a disk axis.
4. No open-source Synology API mock exists; a lightweight shim would be new code.
5. Remaining UNCONFIRMED: rootless viability of vdsm/dockur (tun, `NET_ADMIN`, vhost-net), Synology API PDF parameter details, vsftpd/unfs3/Apache/nginx licences from primary pages, Unraid, whether /dev/sda1 (7.3T) is mounted and free.